(POSIX) Threads

package for man pages: manpages-posix-dev glibc-doc

  sudo apt-get install manpages-posix-dev glibc-doc

setting cores:

  taskset -c 0 <program> -> execute program only with core 0

#include <pthread.h>
gcc/g++ -pthread

defensive programming -> int functions returning success (0) ou failure (!= 0)

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SO library:

Defensive programming removed (NZ macro)

#include <hdp.h>

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THREADS:
opaque type: pthread_t // never compare pthread_t variables

NZ(pthread_create)
NZ(pthread_join)
NZ(pthread_detach)
NZ(pthread_exit)
pthread_equal -> returns int (C boolean)
pthread_self -> returns pthread_t

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MUTEX:
opaque type: pthread_mutex_t

PTHREAD_MUTEX_INITIALIZER
 or
NZ(pthread_mutex_init)
NZ(pthread_mutex_destroy)

NZ(pthread_mutex_lock(mtx))
int not_locked = pthread_mutex_trylock(mtx); NZ(not_locked == EBUSY ? 0 : not_locked);
NZ(pthread_mutex_unlock(mtx))

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CONDITION VARIABLE:
opaque type: pthread_cond_t

PTHREAD_COND_INITIALIZER
 or
NZ(pthread_cond_init)
NZ(pthread_cond_destroy)

NZ(pthread_cond_wait(cvar, mtx))
NZ(pthread_cond_signal(cvar))
NZ(pthread_cond_broadcast(cvar))

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C-Library

errno (shared) global variable, thus not thread-safe!

Beware of Thread-safe functions!

usleep
nanosleep
rand_r